Automated Road Damage Detection Using YOLOv8

Abstract

Timely and accurate assessment of road infrastructure is essential for ensuring transportation safety and efficient urban planning. Traditional manual inspection methods are slow, labor-intensive, and prone to subjectivity, making them unsuitable for large-scale road networks. This project presents an automated road damage detection system using a deep learning–based object detection approach. Leveraging the YOLOv8 architecture and the Road Damage Detection 2022 (RDD2022) dataset, the proposed system detects and classifies multiple categories of road damage from real-world images. The solution demonstrates strong detection performance while maintaining computational efficiency, making it suitable for real-time and large-scale deployment scenarios.

1. Problem Statement

Road surface degradation such as cracks and potholes significantly impacts vehicle safety, travel efficiency, and infrastructure longevity. Existing road inspection practices rely heavily on manual surveys or rule-based image processing techniques, which are costly, time-consuming, and difficult to scale. Additionally, human inspection introduces subjectivity and inconsistency in damage assessment.

The objective of this project is to design a **robust, automated, and scalable computer vision system** capable of detecting and localizing different types of road damage from images. The system should accurately classify each damage instance and provide precise bounding box localization, enabling authorities to make data-driven maintenance decisions.

2. Dataset Description

The proposed solution utilizes the **Road Damage Detection 2022 (RDD2022)** dataset, a large-scale, multi-national dataset specifically curated for road damage analysis under real-world conditions.

a) Dataset Characteristics

- Over **47,000 high-resolution images**
- Images captured from multiple countries, road types, and environmental conditions
- Contains complex scenarios including shadows, varying illumination, and multiple damage instances per image

b) Damage Categories

The dataset includes five classes of road damage:

1. Longitudinal Crack
2. Transverse Crack
3. Alligator Crack
4. Other Corruption
5. Pothole

c) Annotation Format

- Annotations are provided in **YOLO format**
- Each image is associated with a text file containing normalized bounding box coordinates
- This format ensures compatibility with real-time object detection pipelines

d) Data Split

- Training set: Model learning
- Validation set: Performance monitoring and hyperparameter tuning
- Test set: Final prediction generation (labels withheld for evaluation)

3. Model Architecture (YOLOv8m)

This project employs **YOLOv8m**, a modern one-stage object detection model that offers an optimal balance between detection accuracy and inference speed.

Architectural Overview

- **Backbone:** CSP-based convolutional layers for efficient hierarchical feature extraction
- **Neck:** PAN-FPN architecture enabling multi-scale feature fusion for detecting objects of varying sizes
- **Detection Head:** Decoupled head architecture separating classification and localization tasks for improved optimization

  YOLOv8 performs object detection in a single forward pass, making it well-suited for real-time inference and deployment on resource-constrained environments. Pretrained weights (COCO dataset) were used to accelerate convergence and improve generalization.

4. Training Strategy and Data Augmentation

Training Strategy

- Transfer learning was applied using pretrained YOLOv8 weights
- The model was fine-tuned on the RDD2022 dataset to adapt to road-specific visual patterns
- Validation was performed after each epoch to monitor convergence and avoid overfitting

Data Augmentation

To improve robustness under diverse environmental conditions, advanced augmentation techniques were employed:

- Mosaic augmentation for improved object scale variation
- HSV color jittering to handle illumination changes
- Random horizontal flipping
- Gaussian blur to simulate motion and camera noise
- CLAHE for contrast enhancement in low-visibility scenarios

  These augmentations significantly improved the model's ability to generalize to unseen road conditions.

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5. Hyperparameter Configuration

The following hyperparameters were selected based on empirical evaluation and best practices for object detection:

- **Model:** YOLOv8m
- **Input Resolution:** 640 × 640
- **Batch Size:** 16
- **Optimizer:** AdamW
- **Initial Learning Rate:** 0.001
- **Training Epochs:** 20–30
- **Loss Components:**
  - Bounding box regression loss
  - Classification loss
  - Distribution Focal Loss (DFL)

Early stopping was applied to prevent overfitting and reduce unnecessary computation.

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6. Experimental Results and Evaluation

Model performance was evaluated using **Mean Average Precision (mAP)**, the standard metric for object detection tasks.

Evaluation Metrics

- **mAP@0.5:** Measures detection accuracy at an IoU threshold of 0.5
- **mAP@0.5:0.95:** Measures robustness across multiple IoU thresholds

Results

- Achieved **mAP@0.5 in the range of 0.55–0.60** on the validation set
- Demonstrated strong detection accuracy for longitudinal and alligator cracks
- Maintained consistent performance across varying road conditions

These results indicate that the model effectively balances precision and recall while maintaining stable localization accuracy.

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7. Conclusion and Future Work

This project demonstrates the effectiveness of deep learning-based object detection for automated road damage analysis. By leveraging YOLOv8 and a large-scale, real-world dataset, the system provides a scalable and efficient alternative to manual road inspections.

Future enhancements may include:

- Incorporating temporal data from video streams
- Integrating severity estimation for damage prioritization
- Deploying the model on edge devices for real-time monitoring

Overall, the proposed solution presents a practical and industry-ready approach for intelligent infrastructure maintenance.